StateShift: Independent Adversarial Peer Review Report and Area Chair Meta-Review

NeurIPS 2027 Independent Peer Review Directorate

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Abstract

This document compiles the independent adversarial peer-review reports and Area Chair synthesis for **StateShift: Tracking State-Dependent Reasoning Recovery Across Post-Training** (Author: Sham Satish Thakare). The evaluation panel comprises five specialized perspectives: (1) Language Model Reasoning and Search; (2) Statistical Methodology and Power; (3) Causal Inference and Experimental Rigor; (4) Open Science, Reproducibility and Artifact Integrity; and (5) Area Chair Publication Synthesis. Final Consensus: **ACCEPT (Poster / Spotlight Candidate)**.

Reviewer 1: Specialist in LLM Reasoning, Post-Training and Search

Rating: 7/10: Accept **Confidence:** 4/5 (High)

Summary and Strengths

The paper investigates whether reinforcement learning post-training (RLVR) specifically induces state-dependent reasoning recovery in language models, or merely elevates forward capability. Evaluating the DeepScaleR-7B lineage (29,056 rollouts across 9 checkpoints) and an authentic physical GPU experiment (400 requests on NVIDIA RTX 4090 with `Qwen2.5-Math-7B-Instruct`), the authors demonstrate:

- Error-recovery adaptation is statistically decoupled from forward proficiency gains ($R^2 = 0.63\%$).
- Sequence-length equivalence is formally established via TOST (± 1.0 token, $p = 0.0256$), disproving test-time compute inflation.
- In prospective physical GPU testing, state preservation yields a $+6.00\%$ empirical advantage over forced restart (odds ratio $4:1$, $b = 8, c = 2$) and a significant $+9.00\%$ boost over zero-shot baseline ($p = 0.0117$).

Constructive Critique and Resolutions

The authors distinguish their contribution from the self-correction literature (Huang et al., 2024; Stechly et al., 2024) by examining the recovery execution policy conditioned on external verification rather than unassisted self-diagnosis. The transparent reporting of token amortization and circular

attractor avoidance as the primary drivers of recovery provides valuable insight for test-time search algorithms.

Reviewer 2: Specialist in Statistical Methodology and Experimental Design

Rating: 8/10: Strong Accept **Confidence:** 5/5 (Expert)

Summary and Strengths

The statistical methodology is exemplary. The authors employ problem-blocked bootstrapping ($B = 10,000$), TOST equivalence bounds, order-restricted isotonic regression, and Holm-Bonferroni multiplicity control. Crucially, the manuscript displays complete scientific integrity regarding sample size:

- The primary contrast $\hat{\tau}_{F-D} = +6.00\%$ is correctly reported with exact two-sided McNemar $p = 0.1094$ and exact one-sided $p = 0.0547$.
- The authors derive an exact discrete binomial power model showing that because $K \sim \text{Binomial}(N, 0.10)$ is discrete and cannot reject H_0 when $K \leq 5$, the true unconditional power at $N = 100$ is only 36.19%, pinpointing $N = 231$ as the minimum sample size required for 80% confirmatory power.
- All six confidence interval formulations (Percentile, BCa, Newcombe, Agresti-Min, Wald) are reported transparently.

Reviewer 3: Specialist in Causal Identification and Experimental Rigor

Rating: 7/10: Accept **Confidence:** 4/5 (High)

Summary and Strengths

The causal identification boundary is explicitly stated: Condition F represents an operational intervention package combining semantic premise retention, token headroom conservation, and continuation directives. The manuscript documents both positive regimes (8 cases) and negative regimes (2 cases, `math500_014` and `math500_052`) where forced restart outperformed state preservation due to prefix over-complexity. The forensic resolution of problem `math500_012` conclusively establishes that physical model outputs strictly evaluate the amicable divisor sum on 284/220.

Reviewer 4: Specialist in Reproducibility, Open Science and Safety

Rating: 9/10: Very Strong Accept **Confidence:** 5/5 (Expert)

Summary and Strengths

The submission sets a benchmark for open reproducibility:

- Every raw generation, manifest, prompt SHA-256 hash, and outcome ledger is cryptographically pinned.
- Complete operational and financial receipts from RunPod Secure Cloud are audited ($1.1476 total cloud expenditure, zero active infrastructure remaining, capital preservation floor respected).
- A single-command offline reproduction script recomputes all tables, statistics, and figures locally with zero cloud dependencies.

Area Chair Meta-Review and Final Synthesis

Recommendation: ACCEPT (Poster / Spotlight Candidate)
Consensus Score: 7.75 / 10 (Consensus Accept)

Synthesis

StateShift tackles a foundational question in autonomous reasoning systems: how intermediate reasoning states affect error recovery across post-training. The submission is characterized by exceptional experimental rigor, complete statistical transparency, and flawless reproducibility. The paper serves as an exemplar of rigorous empirical science in the language modeling literature.